

Project Proposal – Closet Buddy/Outsy (Viom Shrestha)

1. Project Information

Title: Closet Buddy – Smart Virtual Wardrobe & Sustainable Outfit Recommender

Overview:

Closet Buddy is an AI-powered full-stack system that digitizes a user's wardrobe, classifies clothing items, and recommends sustainable outfits. Users upload clothing images, which are segmented and classified into categories (Sub-Category, Article Type, Usage). Extracted clothes are stored in a virtual closet and used by a hybrid rule-based + AI recommendation engine. The system prevents illogical suggestions (two shirts together), suggests color/style-matched outfits, tracks wear frequency. It also integrates an e-commerce site to suggest wardrobe additions, while promoting sustainability by tracking clothing usage and encouraging reuse, donation, or thrift.

Domain: Artificial Intelligence, Computer Vision, Recommender Systems, Full-Stack Development, Sustainable Fashion

2. R&D on Similar Systems

National (Nepal-based / Local)

- Daraz Nepal (E-commerce platform): Focuses on online fashion sales but lacks personalized outfit recommendations or wardrobe tracking.
- SastoDeal: Offers affordable clothing but does not provide AI-based outfit matching or sustainability tracking.
- HamroBazar (Thrift & Sell Platform): Provides a platform to buy/sell second-hand clothes but without AI-based style recommendations or wardrobe management.

International

- Cladwell (US): A personal styling app that recommends daily outfits but does not include closet digitization via image upload.
- Finery (US): A wardrobe management app that imports online shopping receipts to build closets but lacks AI-based segmentation from photos.
- Combyne (Germany): A fashion app where users manually mix and match clothing for style inspiration, but no automated outfit recommendation engine.

Gap Identified:

Existing systems either focus on fashion shopping, manual outfit creation, or basic styling apps. Very few combine AI-based segmentation, wardrobe management, outfit recommendation, and sustainability tracking into a single platform.

3. Problem Statement

People often struggle to decide what to wear daily or for specific occasions. They may forget what clothes they already own, repeat outfits, or fail to make good style combinations. Current fashion apps either focus on shopping or basic styling but fail to provide an integrated, AI-driven system for closet management, sustainable usage, and intelligent outfit recommendation. There is a need for a system that not only recommends outfits from existing clothes but also reduces fashion waste by promoting reusability and sustainable shopping habits.

4. Proposed Solution

Closet Buddy provides a **virtual closet + AI recommendation system with integrated store**:

- Users upload clothing images then system extracts cutouts, identifies clothing type, and stores them virtually.
- A rule-based + AI-powered recommendation engine suggests color-coordinated, style-appropriate outfits, ensuring logical pairings (no two tops/shoes together).
- The system tracks wear frequency to encourage sustainable fashion choices and reduce overbuying.
- An integrated store recommendation system suggests missing wardrobe items, with options to donate, thrift, or sell clothes.
- User feedback loop improves recommendations over time.

5. Aim & Objectives

Aim:

To develop an AI-driven wardrobe management and fashion assistant that provides outfit recommendations and promotes eco-friendly fashion usage with a built-in clothing store as well.

Objectives:

- Implement an image segmentation model to extract clothing cutouts from uploaded images.
- Extract key features (color, type, category) for each clothing item.
- Design a rule-based outfit recommendation system with category and color harmony rules.
- Extend recommendation engine with AI-based personalization using user preferences and feedback.
- Build a virtual closet to store, browse, and manage outfits and clothes.
- Track clothing usage to encourage reuse and sustainability.
- Integrate a clothing store for wardrobe suggestions, thrift, and donation options.
- Deliver a full-stack platform for end-user interaction.

6. Features to Deliver

- i. Upload clothing images: automated segmentation & classification.
- ii. Virtual closet (browse & manage clothes).
- iii. Outfit recommendation engine (rule-based + AI hybrid).
- iv. Category rule: ensure logical combinations (top + bottom + shoes/accessory).
- v. Outfit feedback loop for personalization.
- vi. Usage tracking: wear history, frequency count.
- vii. Store integration. //Not Fixed
- viii. Sustainability options: thrift, donate, resell. //Not Fixed

7. Platform & Tech Stack

Frontend: React

Backend: Django REST Framework

AI/ML: Python, TensorFlow

Database: PostgreSQL

Deployment: AWS

8. IDE & Tools

IDE: PyCharm?, VS Code

Version Control: Git & GitHub.

Wireframe: Miro

9. Selected Model and Dataset

Segmentation Model: Pretrained U²-Net / Mask R-CNN.

Classification Models:

- prithivMLmods/Fashion-Product-subCategory: fine-grained category classification.
- prithivMLmods/Fashion-Product-articleType: identifies specific clothing type (hoodie, sandals, etc).
- prithivMLmods/Fashion-Product-Usage: predicts clothing usage context (casual, formal, ethnic, party).

Color Extraction: K-means clustering.

Recommendation Logic:

- Rule-based: category rules + color theory + context filtering.
- AI-based: convert item metadata into text descriptions to embeddings (e.g., Sentence-BERT / CLIP text encoder) to similarity search for outfit matching.

Dataset:

- DeepFashion2 Dataset.
- Custom dataset for fine-tuning or evaluation.